

$$\frac{9}{9} \Rightarrow \frac{100}{100} \%$$

0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	100
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Name:

These questions assess your understanding of the material from Chapters 20, 21 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. An incandescent lightbulb has a 1.4-cm-long filament and operates at 108 V to deliver 36 W. Calculate the diameter of the bulb's filament, assuming it to be a cylindrical tungsten wire, given that the resistivity of tungsten is $5.6 \times 10^{-8} \Omega \cdot \text{m}$.

$$R = \frac{\rho L}{\pi r^2} \Rightarrow d = 2 \sqrt{\frac{\rho L}{\pi R}}$$

$$\text{ANSWER: } 1.755 \times 10^{-6} \text{ m}$$

$$R = \frac{V^2}{P} = 324 \Omega$$

$$d = \sqrt{\frac{(5.6 \times 10^{-8})(0.014)}{\pi (324)}} \times 2$$

$$= 1.755 \times 10^{-6} \text{ m}$$

2. A 0.62-m-long copper wire of diameter 0.50 mm is connected to a 6.7-V battery at 8.4°C. Calculate the percent change in the power delivered by the wire when it is heated to 51°C. The resistivity of copper at 8.4°C is $1.7 \times 10^{-8} \Omega \cdot \text{m}$ and its linear thermal expansion coefficient is $3.9 \times 10^{-3} \frac{1}{^\circ\text{C}}$.

$$\rho(T) = \rho_0 (1 + \alpha \Delta T) \quad \Delta T = 51 - 8.4 = 42.6^\circ\text{C}$$

$$\text{ANSWER: } 14.25\% \text{ Loss}$$

$$\rho(T) = (1.7 \times 10^{-8})(1 + (3.9 \times 10^{-3})(42.6)) = 1.9824 \times 10^{-8} \Omega \cdot \text{m}$$

$$P_1 = \frac{V^2}{R_1}, \quad R_1 = \frac{\rho_1 L}{A} = \frac{(1.7 \times 10^{-8})(0.62)}{\pi (0.25 \times 10^{-3})^2} = 0.05368 \Omega \Rightarrow P_1 = 836.2519 \text{ W}$$

$$R_2 = \frac{(1.9824 \times 10^{-8})(0.62)}{\pi (0.25 \times 10^{-3})^2} = 0.06280 \Omega \Rightarrow P_2 = 717.0927$$

$$\Delta\% = 1 - \frac{P_2}{P_1} = 14.25\%$$

3. Suppose the voltage supplied to a circuit is 12 V. There are three resistors wired in series: $R_1 = 2.7 \Omega$, $R_2 = 3.5 \Omega$, and $R_3 = 2.3 \Omega$. Calculate the current in the circuit.

$$V = IR \Rightarrow \frac{V}{R} = I$$

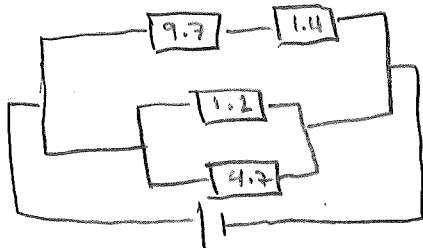
$$\text{ANSWER: } 1.412 \text{ A}$$

$$R_s = R_1 + R_2 + R_3 = 8.5 \Omega$$

$$I = \frac{12 \text{ V}}{8.5 \Omega} = 1.412 \text{ A}$$

+3

4. A circuit has two branches connected in parallel: branch A consists of resistors $R_1 = 9.7\Omega$ and $R_2 = 1.4\Omega$ in series; branch B consists of resistors $R_3 = 1.2\Omega$ and $R_4 = 4.7\Omega$ in parallel. Draw this circuit and find the total equivalent resistance. Note: you must draw the circuit correctly to receive credit.



$$R_A = 9.7\Omega + 1.4\Omega \quad \text{ANSWER: } 0.88\Omega$$

$$R_B = \left(\frac{1}{1.2} + \frac{1}{4.7}\right)^{-1}$$

$$R_T = \left(\frac{1}{R_A} + \frac{1}{R_B}\right)^{-1}$$

$$R_A = 11.1 \quad R_B = 0.9559$$

$$\Rightarrow R_T = 0.88$$

5. What current flows through a high-voltage transmission line to transmit 260MW of power at 240kV?

$$\text{ANSWER: } 1.08 \times 10^3 \text{ A}$$

$$P = IV \Rightarrow I = \frac{P}{V}$$

$$= \frac{260 \times 10^6 \text{ W}}{240 \times 10^3 \text{ V}} =$$

6. Calculate the power dissipated by a car headlight that draws a current of 4.0A when it has a voltage applied to it of 14. V.

$$\text{ANSWER: } 56 \text{ W}$$

$$P = IV = 4 \text{ A} \times 14 \text{ V} = 56 \text{ W}$$

7. A 1.1-m-long copper wire must carry 14. A without dissipating more than 11. W as heat. Calculate the wire's minimum diameter given that copper's resistivity is $1.7 \times 10^{-8} \Omega \cdot \text{m}$.

$$d = 2 \sqrt{\frac{\rho L}{\pi R}}$$

$$\frac{P}{I^2} = R = \frac{11}{14^2} = 0.05612 \Omega$$

$$\text{ANSWER: } 6.514 \times 10^{-4} \text{ m}$$

$$d = 2 \sqrt{\frac{(1.7 \times 10^{-8})(1.1)}{(0.05612)(\pi)}} = 6.514 \times 10^{-4} \text{ m}$$

+4

8. What is the value of the peak voltage for 280-V AC power?

$$V_{RMS} = \frac{V_0}{\sqrt{2}} \Rightarrow V_{RMS} \sqrt{2} = V_0$$
$$= 280 \sqrt{2} = 395.98 \text{ V}$$

ANSWER: 395.98 V



9. A car headlight filament is made of tungsten ($\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$) and has a resistance of 0.11Ω . If the filament is a cylinder 1.2-cm long, what is its diameter?

$$d = 2 \sqrt{\frac{\rho L}{\pi R}} = 2 \sqrt{\frac{(5.6 \times 10^{-8})(0.012)}{(0.11) \pi}}$$
$$\Rightarrow 8.819 \times 10^{-5}$$

ANSWER: $8.819 \times 10^{-5} \text{ m}$



+2

1. $d = 1.76 \times 10^{-6} \text{ m} \therefore P = \frac{V^2}{R}, R = \frac{\rho L}{A} \text{ and } A = \pi r^2$
2. $\% \Delta P = -14.2\% \therefore \rho(T) = \rho_0(1 + \alpha \Delta T), R = \frac{\rho L}{A}, P = \frac{V^2}{R}$
3. $I = 1.41 \text{ A} \therefore R_s = R_1 + R_2 + R_3, V = IR$
4. $R_{eq} = 0.880 \Omega \therefore R_s = R_1 + R_2 + R_3, \frac{1}{R_p} = \frac{1}{R_1} + \dots + \frac{1}{R_n}$
5. $I_{RMS} = 1.08 \times 10^3 \text{ A} \therefore P_{ave} = I_{RMS} V_{RMS}$
6. $P = 56.0 \text{ W} \therefore P = IV$
7. $d_{min} = 6.51 \times 10^{-4} \text{ m} \therefore R = \frac{\rho L}{A}, P_{int} = I^2 R$
8. $V_0 = 396. \text{ V} \therefore V_{RMS} = \frac{V_0}{\sqrt{2}}$
9. $D = 8.82 \times 10^{-5} \text{ m} \therefore R = \frac{\rho L}{A}$